

## Industry experts' suggestions

Analogue circuit design is often considered a 'mystical art.' It requires long practice to become proficient in the dark art of analogue design. The connection between cause and effect is vague with so many physical and electrical interactions between components. A weak ground and some stray capacitance can make designs falter. The PCB layout also weights to a greater part of the analogue equations.

Analogue design needs a totally different approach to solving problems. It makes good sense to rig up the circuit and test, to unravel the mystery associated with a problem. Don't use simulation tools (Spice, Saber, etc) to do your design, but only to confirm your design when in doubt or perhaps to do a detailed analysis (tolerance, temperature, etc). Do not blindly copy application circuits from datasheets. If you follow these generic rules, you will be able to break the myth of 'analogue circuit design is black magic.'

—**Basavaraj Garadi, Chief Expert, Robert Bosch Engineering and Business Solutions**

1. Keep up-to-date with the basics of analogue and linear circuits.
2. Visualise any circuit you test to the transistor level.
3. Take up your own project and materialise it. This will give you exposure to the real world of analogue design.
4. Try making circuits that work with small signals, sensors and instrumentation. These will let you experience the noise and start working on eliminating it.
5. Become proficient in building and testing an analogue electronic circuit.

—**P. Chow Reddy, manager-R&D (Power Division), ICOMM Tele Ltd**

As analogue skills are more complex than other VLSI skills, many engineers tend to choose career in easier domains. However, many bright engineers who chose jobs in other domains after graduation, are considering to switch to analogue domain as their current job is no longer challenging. Hence I would suggest bright engineers to choose analogue domain and not just any job after their graduation.

—**Somashekhar B., technology manager-analogue/mixed-signal group, Semicon Technology-Systems BU, Tata Elxsi**

A large number of MNCs have invested heavily in analogue design in their India centres. I also see many niche start-up companies emerging in this space. They plan to develop analogue IP for licensing as well as provide design services. Academia is also acting as a catalyst as many research projects are ongoing in niche analogue areas like power management and high-speed SerDes (serialiser/deserialiser) designs."

Somashekhar B., technology manager-analogue/mixed-signal group, Semicon Technology-Systems BU, Tata Elxsi, adds, "More of analogue/mixed-signal design work is being outsourced to India, which was earlier done within the semicon product companies as core technology. As there is acute shortage of analogue resources globally, India is seen as one of the preferred countries for analogue design."

### Analogue scope in a digital world

According to Reddy, analogue circuit design requires more skills than digital

systems design. An analogue circuit must be designed by hand, and the process is much less automated than digital systems.

Reddy says, "A design engineer having strong understanding of analogue electronics and capable of handling analogue components, small signals and the sensor to the instrumentation will always sit in a high profile having scope to work in power, digital, RF and instrumentation," he adds.

As far as board designs are concerned, Garadi believes that pure analogue circuit design does not have significant scope at the moment.

"At least not in the league of what used to happen in the good old 80's and the 90's. Back then, circuit diagrams would span over several sheets without you encountering a single microcontroller or a logic circuit. But now, the closest that one would be doing is mixed-signal circuit design. On that count, most boards interacting with the real world would certainly have front-end interfaces involving signal conditioning-amplifying, attenuating, fil-

tering or buffering signals," says Garadi.

However, Garadi believes IC, ASIC or VLSI designs are still the areas of interest to hardcore analogue designers, and look very promising in terms of salaries. Almost all the big names in semiconductor industry—STMicroelectronics, Texas Instruments, Freescale Semiconductors and Maxim Integrated to name a few—have offices or design centres in India. This sector provides fairly decent opportunities either as an IC designer or as an application engineer for linear ICs.

According to Rajani Rao, for a major semiconductor company to succeed, it has to have a strong analogue design team.

Attributing to the fact that analogue design is indeed more challenging than digital, Somashekhar says, "As analogue skills are more complex and unique, it takes slightly more time to ramp-up. But once a person attains that level, it leads to a stable, long-term and highly rewarding career growth."

"The design tasks are highly challenging and provide great job satisfaction. It is a pride to claim that you are an analogue engineer—a rare skill to stand out among others," he adds.

### Qualification

"For analogue board design, a bachelor's degree in electronics or one of its offshoots like electronics & communication or instrumentation would do. However, for engineers wanting to pursue a career in IC or VLSI design, a master's programme in microelectronics and VLSI design or its equivalent would be the minimum requirement," says Garadi.

"A master's or above degree in integrated electronics is the most desirable. However, nowadays there is exposure to analogue design even at the undergraduate level," adds Rajani Rao.

Dr S. Karthik, engineering director, India Product Development Centre (IPDC), Analog Devices India, too feels that for analogue design, an M.Tech in VLSI would be preferred. However, according to him, "B.Tech candidates with a strong background and interest in analogue design would also do well."